Amazon Web Services: Amazon Simple Storage Service (S3)

Topics

- What is S3?
- S3 Basics
- Working with S3 Buckets
- Key-Value Store
- Availability and Durability
- Characteristics
- Securing Your Data
- Consistency Model
- Storage Classes
- Demos

What is S3?

- Stands for Simple Storage Service
- Object storage in the cloud
- Provides secure, durable, and highly scalable object storage
- Simple to use
- Manages data as objects rather than in file systems or data blocks

Amazon Simple Storage Service
(Amazon S3)

S3 Basics

Unlimited Storage

The total volume of data and the number of objects you can store is unlimited

Objects up to 5 TB in size

S3 objects can range in size from a minimum of 0 bytes to a maximum of 5 terabytes

S3 Buckets

Store files in buckets (~ folders)

Bucket

Working with S3

- Universal Namespace
  - All AWS accounts share the S3 namespace.
  - Each S3 bucket name is globally unique
- Example S3 URLs
  - https://bucket-name.s3.Region.amazonaws.com/key-name
  - https://cs7346smu2023.s3.us-east-1.amazonaws.com/test.txt
- Uploading Files
  - When you upload a file to an S3 bucket, the server will respond with an HTTP 200 code if the upload was successful

Key-Value Store

Key

The name of the object (example: test.txt)

Value

The data itself (sequence of bytes)

Version ID

Important for storing multiple versions of the same object

Metadata

Data about the data you are storing (e.g., content-type, last-modified, etc.)

Availability and Durability

S3 is a Safe Place

Data is always spread across multiple devices and facilities

Built for Availability

Built for 99.95% – 99.99% service availability, depending on the S3 tier

Designed for Durability

Designed for 99.999999999% (11 9s) durability for data stored in standard S3

What is Standard S3?

- High Availability and Durability
  - Data is stored redundantly across multiple devices in multiple facilities (>= 3 AZs):
    - 99.99% Availability
    - 99.999999999% (11 9s) Durability
- Designed for Frequent Access
  - Perfect for frequently accessed data
- Suitable for Most Workloads
  - Default storage class
  - Use cases include websites, content distribution, mobile and gaming apps, and big data analytics

Characteristics

- Tiered Storage
  - Range of storage classes designed for different use cases
- Lifecycle Management
  - Define rules to automatically transition objects to a cheaper storage tier or delete objects that are no longer required after a set time
- Versioning
  - All versions of an object are stored and can be retrieved, including deleted objects

Securing Your Data

- Server-side encryption
  - Set default encryption on a bucket
- Access Control List (ACLs)
  - Define which AWS accounts or groups are granted access to buckets/objects
- Bucket Policies
  - Specify what actions are allowed or denied on a bucket

Consistency Model

- Strong read-after-write consistency for PUT and DELETE requests of objects in a bucket in all AWS regions
  - Applies to both writes to new objects as well as PUT requests that overwrite existing objects and DELETE requests
- Strong consistency for any read (GET or LIST requests)
- Updates to a single key are atomic

https://docs.aws.amazon.com/AmazonS3/latest/userguide/Welcome.html#ConsistencyModel

Storage Classes

- **Frequently accessed objects**
  - S3 Standard
  - Reduced Redundancy
- **Automatically optimizing data with changing or unknown access patterns**
  - S3 Intelligent-Tiering
- **Infrequently accessed objects**
  - S3 Standard-IA
  - S3 One Zone-IA
- **Archiving Objects**
  - **S3 Glacier Instant Retrieval**
  - **S3 Glacier Flexible Retrieval**
  - **S3 Glacier Deep Archive**

https://docs.aws.amazon.com/AmazonS3/latest/userguide/storage-class-intro.html

S3 Standard

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S3 Standard

S3 Standard-Infrequent Access

Rapid Access

Used for data that is accessed less frequently but required rapid access when needed.

Pay to Access the Data

There is a low per-GB storage price and a per-GB retrieval fee.

Use Cases

Great for long-term storage, backups, and as a data store for disaster recovery files.

S3 Standard-IA

99.9% Availability
99.999999999% (11 9s) Durability

S3 One Zone-Infrequent Access

- Like S3 Standard-IA, but data is stored redundantly within a single AZ
    - 99.5% Availability
    - 99.999999999% (11 9s) Durability
- Costs 20% less than regular S3 Standard-IA
- Great for long-lived, infrequently accessed, non-critical data

S3 One Zone-IA

S3 Intelligent-Tiering

- 2 Tiers: Frequent and Infrequent Access
  - Automatically moves your data to the most cost-effective tier based in how frequently you access each object
- Data is stored redundantly across multiple devices in multiple facilities (>= 3 AZs):
  - 99.99% Availability
  - 99.999999999% (11 9s) Durability
- Monthly fee of $0.0025 per 1000 objects
- Use case: unknown or unpredictable access patterns

S3 Intelligent-Tiering

Glacier Options

- Glacier Instant Retrieval
  - Provides long-term data archiving with instant retrieval time for your data
- Glacier Flexible Retrieval
  - Ideal storage class for archive data that does not require immediate access but needs the flexibility to retrieve large sets of data at no cost
  - Can be minutes to 12 hours
- Glacier Deep Archive
  - Cheapest storage class
  - Designed for customers that retain data sets for 7-10 years or longer to meet customer needs and regulatory compliance requirements
  - Standard retrieval time is 12 hours; bulk retrieval time is 48 hours

99.99% Availability
99.999999999% (11 9s) Durability

Demo: Create an S3 Bucket

Object ACLs vs. Bucket Policies

ACL

Object ACLs

Work on an **individual object** level

VS

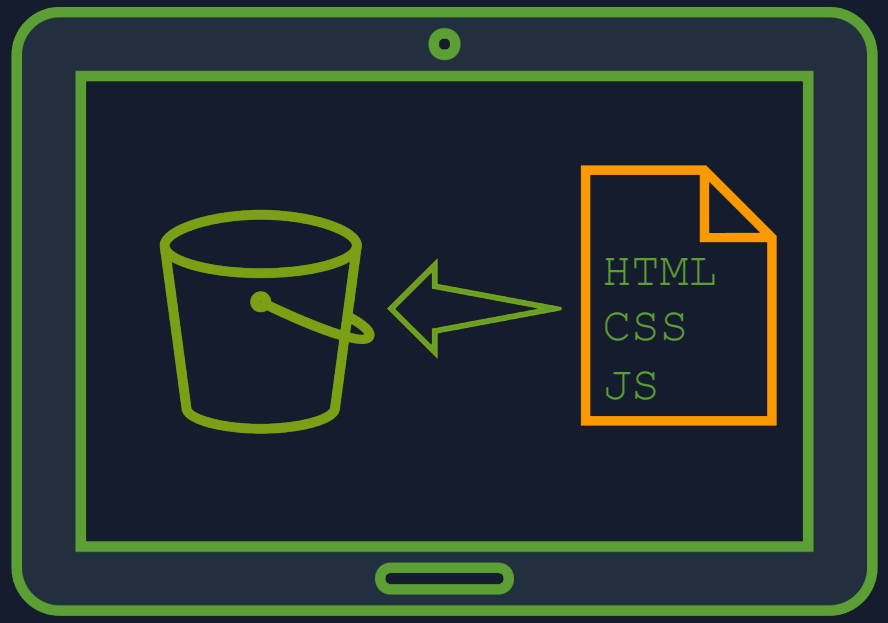
Bucket Policy

Bucket Policy

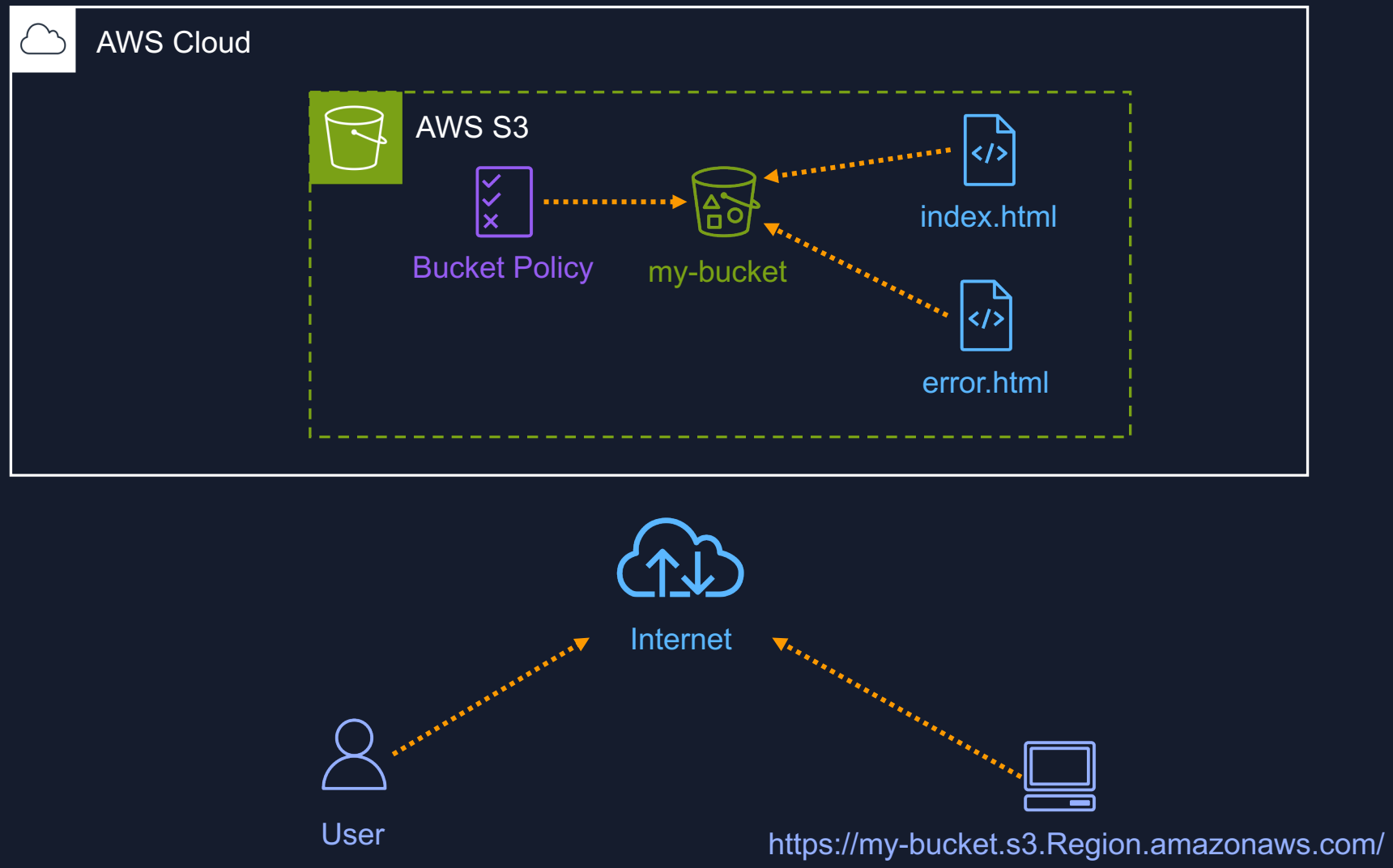
Work on an **entire bucket** level

Demo: Hosting a Static Website

- You can use S3 to host static websites, such as .html sites
- Dynamic websites cannot be hosted on S3
- S3 scales automatically to meet demand



Demo: Hosting a Static Website



Demo: Versioning

- You can enable versioning in S3 so you can have multiple versions of an object within S3
- Advantages of Versioning:
  - All versions of an object are stored in S3
  - Can be a great backup tool
  - Once enabled, cannot be disabled – only suspended
  - Can be integrated with lifecycle rules
  - Can support MFA

Demo: Life Cycle Management

- Automates moving your objects between different storage tiers to maximize cost effectiveness

- Combine with Versioning
  - Move different versions of object to different storage tiers
  - Can be applied to current versions and previous versions of objects

S3 Object Lock

- Used to store objects using a write once, read many (WORM) model
  - Prevent objects from being deleted or modified for a fixed amount of time or indefinitely
  - Can be applied on individual objects or across a bucket
  - Meet regulatory requirements that require WORM storage
  - Add an extra layer of protection against object changes and deletions

S3 Object Lock

S3 Object Lock

- Two lock modes:
  - Governance Mode
    - User need special permissions to overwrite or delete an object version or alter its lock settings
  - Compliance Mode
    - Users, including root, cannot overwrite or delete a protected object version
- A retention period protects an object version for a fixed amount of time
- A legal hold prevents an object version from being overwritten or deleted
  - s3:PutObjectLegalHold permission

Legal hold

Demo: Encrypting S3 Objects

- Types of Encryption
  - Encryption in Transit: SSL/TLS, HTTPS
  - Encryption at Rest: Server-side Encryption
    - SSE-S3: S3-managed keys, using AES256-bit encryption
    - SSE-KMS: AWS Key Management Service – managed keys
    - SSE-C: Customer provided keys
  - Encryption at Rest: Client-side Encryption
    - Encrypt file before uploading them to S3

Data encryption key

Demo: Encrypting S3 Objects

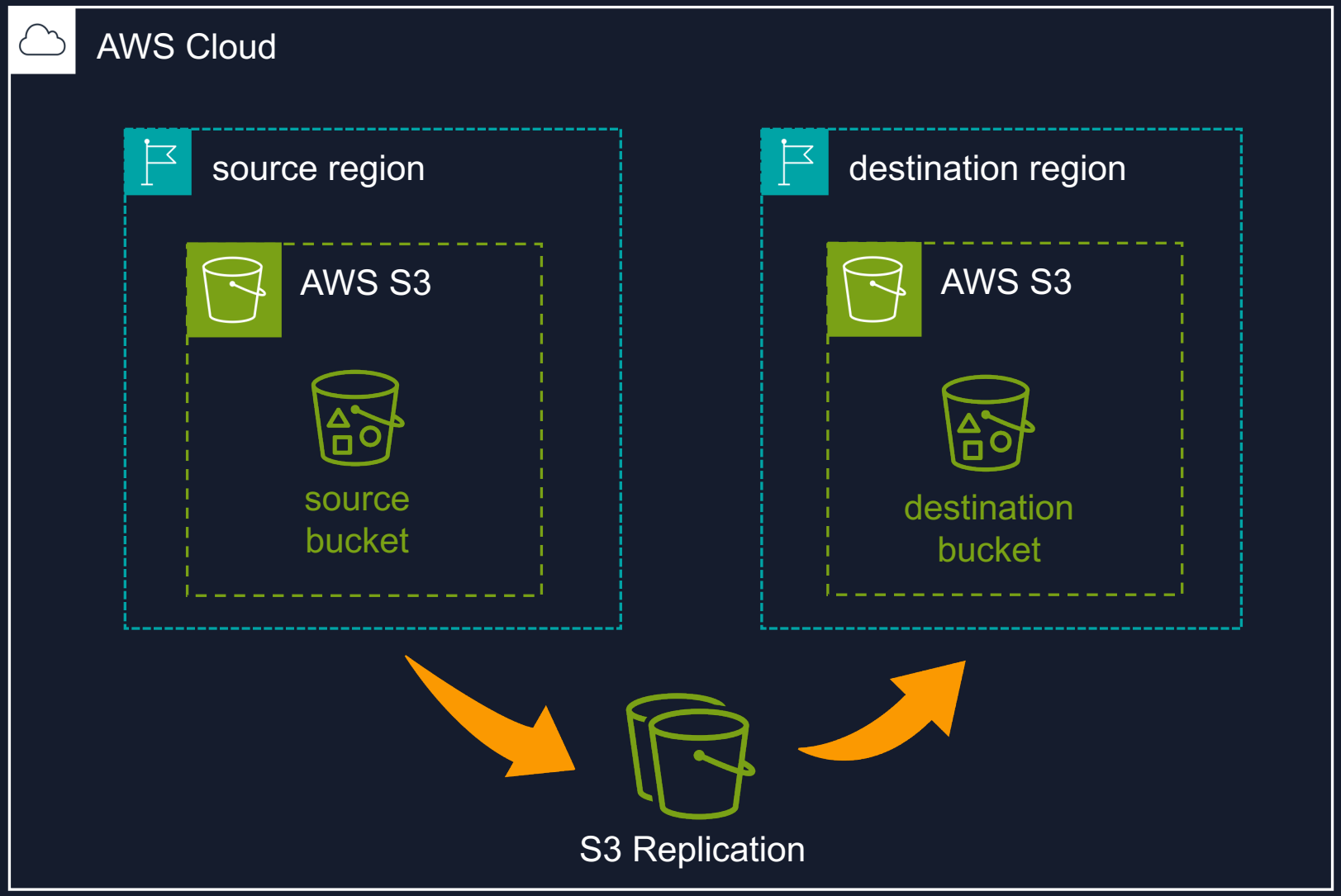
- Server-side Encryption: Enabled by Default
- All Amazon S3 buckets have encryption configured by default
- All objects are automatically encrypted using server-side encryption with Amazon S3 managed keys (SSE-S3)

Demo: S3 Replication

- You can replicate objects from one bucket to another
  - Versioning must be enabled on both the source and destination buckets
- Objects in an existing bucket are not replicated automatically
  - Once replication is turned on, all subsequent updated objects will be replicated automatically

S3 Replication

Demo: S3 Replication



Checkpoint

Your boss instructs you to make your company's S3 buckets public so employees can download content when they are at home. What do you do?

1. Tell your boss to bring a floppy disk to transport the files to and from the office, instead.
2. Open them up! They said it was ok, right?
3. S3 buckets are public by default, so you don't need to do anything to allow this.
4. Explain to your boss that this is a terrible idea and opens a huge security risks for your company.

What is the smallest object you can store in S3?

1. 1 GB
2. 1 MB
3. 0 bytes
4. 1 KB

What kind of data should be stored in S3 Standard?

1. Data that is a secondary copy of a primary source
2. Data that is frequently used
3. Data that is infrequently used
4. Data that is rarely used

Since S3 is an object-based storage solution, which type of file should never be stored in it?

1. Application logs
2. HTML documents
3. Operating system's boot files
4. Cat photos

Why is versioning not enabled by default on new S3 buckets?

1. It slows down the retrieval times in S3.
2. Amazon is mean and wants us to accidentally overwrite our data.
3. It's a security risk, since versioning requires that your content is publicly available.
4. It costs money, as you'll be paying for every additional copy of your objects that you upload.

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